CSC2042S Assignment 1 ~ Adam Foflonker

Preamble

3rdParty Libraries Used

- NumPy
- Pandas
- Scikit-Learn
- Matplotlib

Running the Notebook

Prerequisites for running `notebook-assignment1.ipynb`:

- Aforementioned 3rdParty libraries must be installed.
- The **WDICSV.csv** dataset must be placed in `"data/WDICSV.csv"` - relative to `notebook-assignment1.ipynb`. Note that the `"data/"` directory has been ignored by git.

Global Constants, State and Flags

```
DATA_PATH: Path = Path("./data/WDICSV.csv")
RANDOM_STATE: int = 42

plt.rcParams.update({ "figure.figsize": (8, 5), "axes.grid": True })
np.random.seed(RANDOM_STATE)

# Feature name constants
FEATURE_BASE_COUNTRY_NAME: str = "Country Name"
FEATURE_BASE_COUNTRY_CODE: str = "Country Code"
FEATURE_BASE_INDICATOR_NAME: str = "Indicator Name"
FEATURE_BASE_INDICATOR_CODE: str = "Indicator Code"

FEATURE_BASE_COUNTRY_ALL: list[str] = [FEATURE_BASE_COUNTRY_NAME,
FEATURE_BASE_COUNTRY_CODE]
FEATURE_BASE_INDICATOR_ALL: list[str] = [FEATURE_BASE_INDICATOR_NAME,
FEATURE_BASE_INDICATOR_CODE]
FEATURE_BASE_ALL: list[str] = FEATURE_BASE_COUNTRY_ALL +
FEATURE_BASE_INDICATOR_ALL

FEATURE_DERIVED_YEAR: str = "Year"

# Feature filtering thresholds
MIN_FEATURE_COVERAGE: int = 0.60  # Keep features with >= 60% (non-NaN)
MIN_SAMPLE_COVERAGE: int = 0.70   # Keep samples (country-year) with >= 70% (non-
NaN)

# The best K-value to use for clustering as determined by the silhouette score in
```

```
section 6
BEST_K: int = 4
```

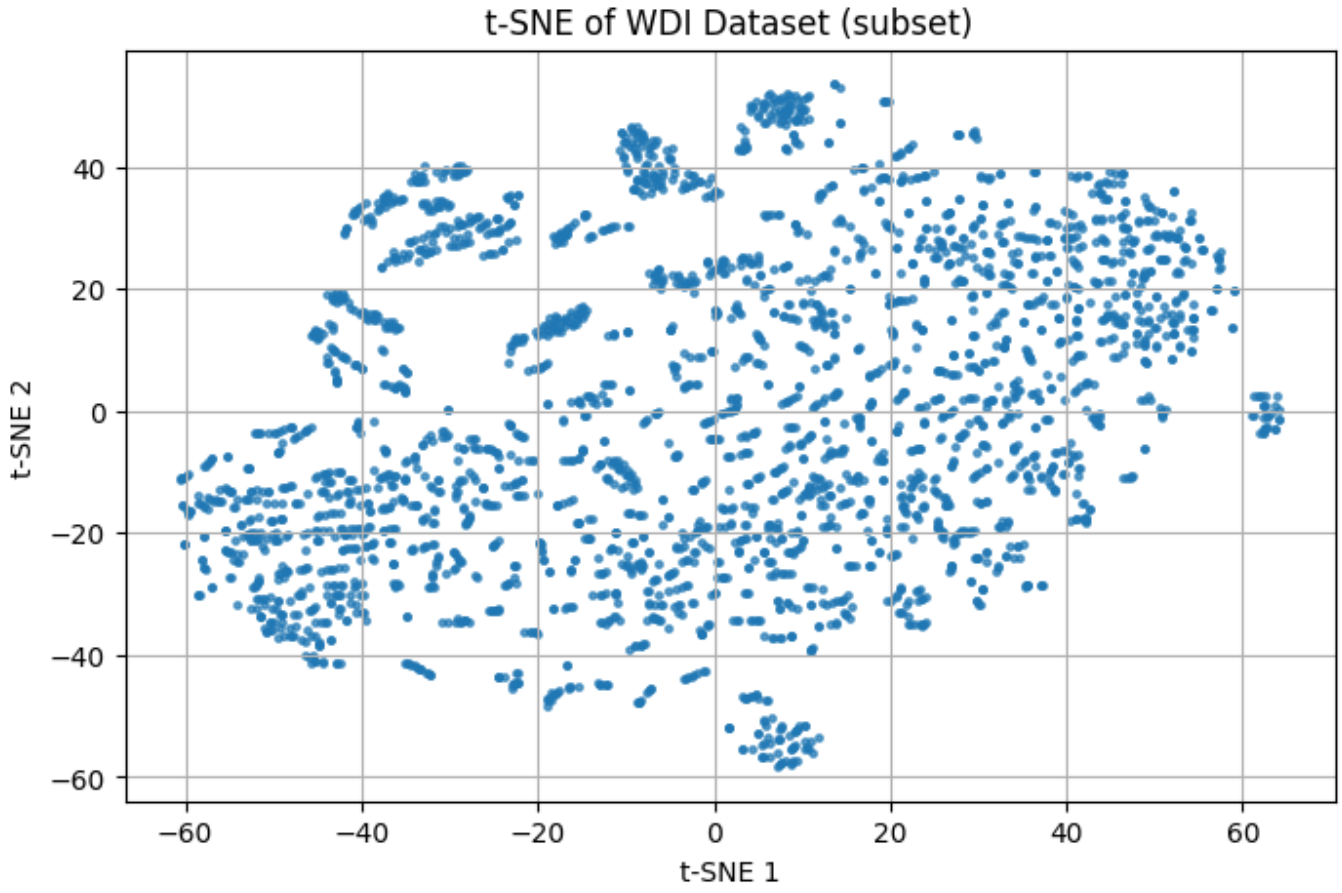
Machine Learning Pipeline Tasks

Data Preprocessing

For preprocessing, all features with greater than `60%` coverage were kept, and all samples with greater than `70%` coverage were kept as well. The rest were either discarded or computed using *median imputation*. The **median** was used instead of the *mean* as it works better for outliers and in general practice in the real world. Scaling was applied to all data points using `sklearn.preprocessing`.

Visualization with t-SNE

Visualization was done using t-SNE. Since the input data set is quite large, only a **subset** of the data points were visualized, as seen below:



K-Means Clustering

A custom implementation of **K-Means Clustering** was implemented using **NumPy**. Both random initialization and **K-Means++** were implemented however, only **K-Means++** was used for the final clustering.

Convergence Criteria

4 Different convergence criterions were tested, namely:

- `max_iters_only`: Tests the convergence purely against the `max_iters` parameter.
- `centroid_tol`: Tests the convergence against the centroid shift tolerance.
- `inertia_tol`: Tests the convergence for inertia tolerance.
- `early_patience`: Tests the convergence under early-stopping conditions.

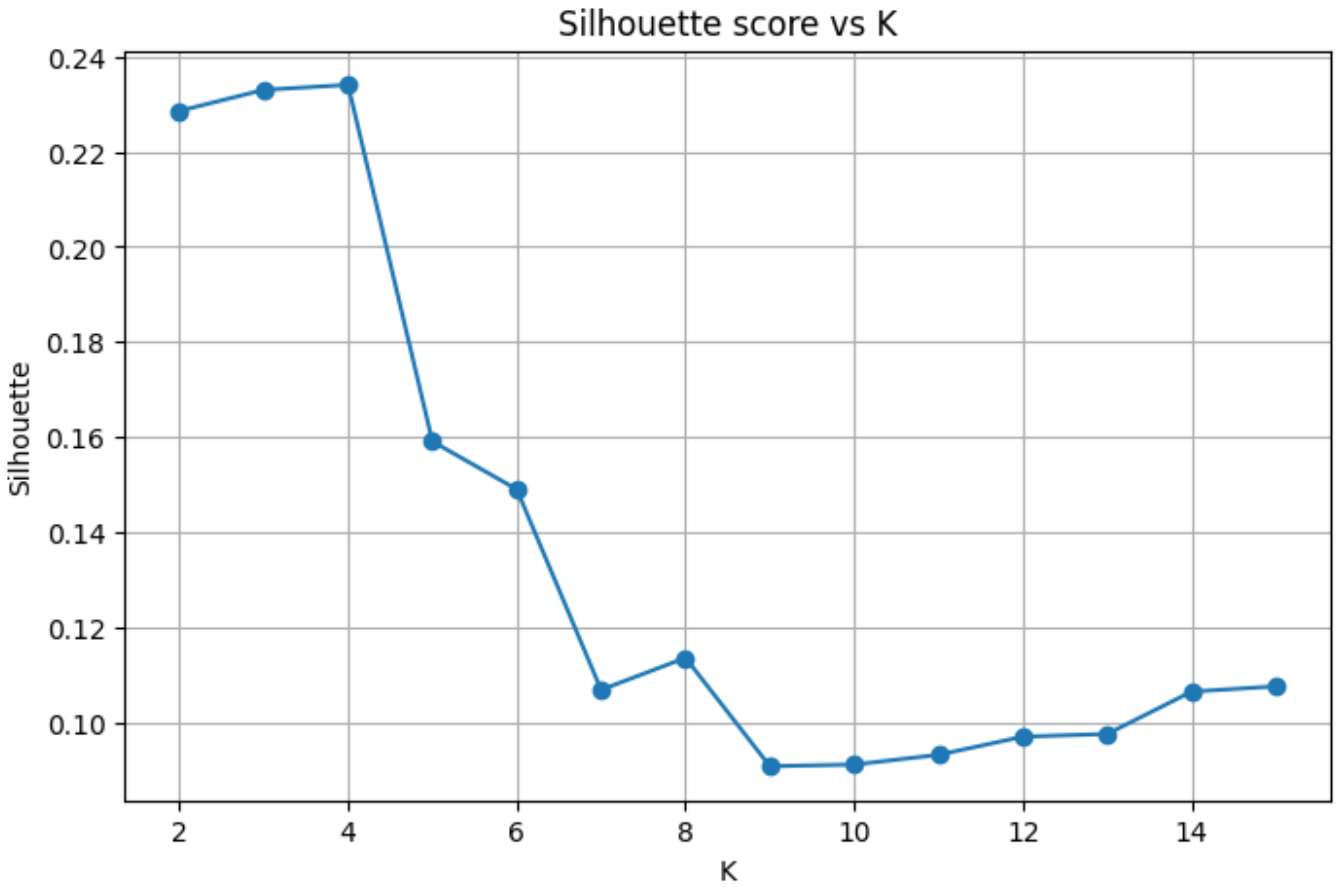
The printed results of these convergence tests are as follows:

```
max_iters_only -> iters: 40 , inertia: 1864980.45
centroid_tol -> iters: 40 , inertia: 1864980.45
inertia_tol -> iters: 40 , inertia: 1864980.45
early_patience -> iters: 40 , inertia: 1864980.45
```

Determining the Optimal-K

Determining the **optimal-K** was done by plotting an *elbow* and a *silhouette* plot. From the *silhouette* plot in particular, it was determined that the **optimal-K** value is 4.

Silhouette score plot:

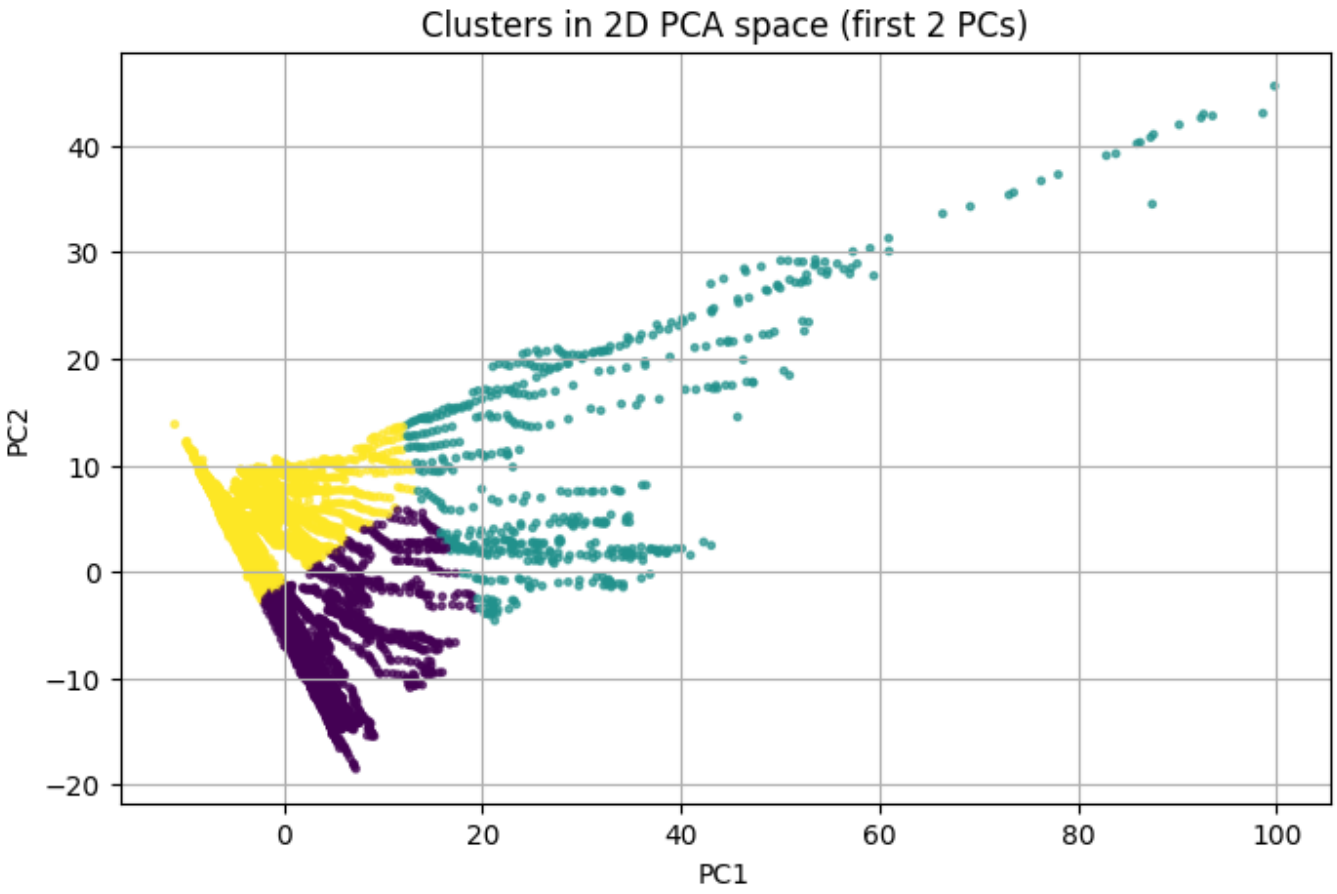


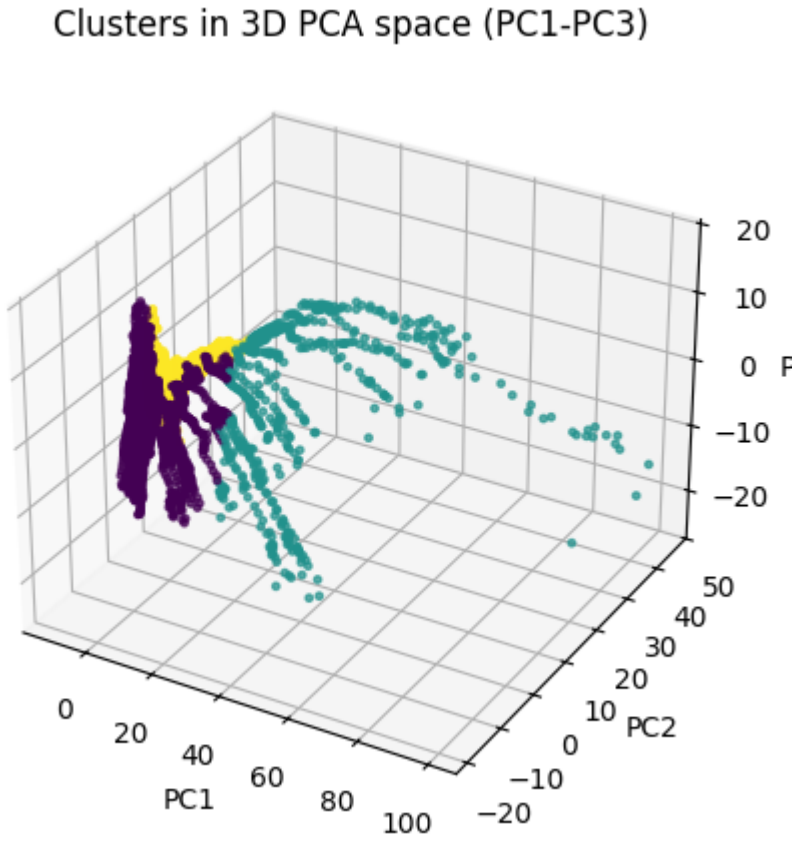
Dimensionality Reduction

Dimensionality reduction was done using **PCA** - configured to use 14 principal components, as determined optimal by the **elbow curve plot** seen earlier. From **PCA**, we are able to view the top features for each **principal component**:

```
Explained variance ratio (first 14): [0.2742 0.2049 0.0516 0.0396 0.0269 0.0238
0.0213 0.0188 0.0156 0.0146
 0.0138 0.0112 0.0109 0.0101]
Top features for PC1:
   SP.POP.65UP.FE.IN  |  Population ages 65 and above, female  | weight= 0.1087
   SP.POP.65UP.TO  |  Population ages 65 and above, total  | weight= 0.1084
   EN.GHG.CO2.MT.CE.AR5  |  Carbon dioxide (CO2) emissions (total) excluding
LULUCF (Mt CO2e)  | weight= 0.1081
   EN.GHG.ALL.MT.CE.AR5  |  Total greenhouse gas emissions excluding LULUCF (Mt
CO2e)  | weight= 0.1081
    ...
Top features for PC2:
   SP.POP.0014.TO.ZS  |  Population ages 0-14 (% of total population)  | weight=
0.1131
   SP.POP.0014.MA.ZS  |  Population ages 0-14, male (% of male population)  |
weight= 0.1128
   SP.POP.0014.FE.ZS  |  Population ages 0-14, female (% of female population)  |
weight= 0.1118
   SP.POP.0509.MA.5Y  |  Population ages 05-09, male (% of male population)  |
weight= 0.1114
   ...
Top features for PC3:
   SP.POP.2529.FE.5Y  |  Population ages 25-29, female (% of female population)  |
weight= 0.1968
   SP.POP.2529.MA.5Y  |  Population ages 25-29, male (% of male population)  |
weight= 0.1946
   SP.DYN.CDRT.IN  |  Death rate, crude (per 1,000 people)  | weight= -0.1682
   SP.POP.3034.FE.5Y  |  Population ages 30-34, female (% of female population)  |
weight= 0.1597
   ...
Inertia (original): 2083608.43
Inertia (PCA 6D): 886916.45
```

We can then view the clusters in both **2D** and **3D** space:





Well-defined and grouped clusters can be observed from these plots.

Cluster Interpretation

On interpretation of these clusters, **cluster representatives** were determined:

```
Cluster 0 representatives:
  Bahamas, The (BHS) - 2013
  New Caledonia (NCL) - 2017
  Bahamas, The (BHS) - 2014
  ...
Cluster 1 representatives:
  Upper middle income (UMC) - 2016
  Upper middle income (UMC) - 2014
  Upper middle income (UMC) - 2015
  ...
Cluster 2 representatives:
  Thailand (THA) - 1977
  Mongolia (MNG) - 1988
  Mongolia (MNG) - 1986
  ...
```

References

- ChatGPT: Debugging, pipeline overview, results/output verification.